# [TSD] AI-Driven Job Matching Platform for Palestine

## Table of Contents



**Purpose:** This document provides a comprehensive description of the technical specifications, configurations, and behaviours of the system or component titled AI-Driven Job Matching Platform. The objective is to ensure clarity for all stakeholders involved in the development, implementation, and maintenance of the system.

# 1. Project Overview

**System Name:** Artificial Intelligence Driven Job Matching Platform for Palestine

**Version:** 1.0

**Author:** Ministry of Labor, Palestinian Employment Fund, and Enabel

**Date:** May 19th 2025

**Description:** The Artificial Intelligence Driven Job Matching Platform is a highly sophisticated standalone internet accessible system engineered to fundamentally address the structural challenges of labour market efficiency in Palestine. Designed as a core component of the Youth Economic Empowerment in Palestine program, the software aims to drastically enhance the employability and economic empowerment of Palestinian youth. This platform specifically targets individuals between fifteen to twenty-nine years old, including vulnerable populations, people with disabilities, and young women. In its future operational state, this software will serve as the central nucleus of a much broader Labor Market Information System, seamlessly integrating multiple external job sources and highly secure government databases to optimize employment matching, generate actionable analytics, and support evidence-based decision making for regional policymakers.

Crucially, due to its status as critical national socioeconomic infrastructure and its direct, real-time integration with sensitive Palestinian Ministry of Labor databases, the platform is strictly classified and engineered to meet the most rigorous Open Worldwide Application Security Project mandates. This absolute highest tier of security classification is mandated because the platform processes highly sensitive personal profiles, verifies official government identification numbers, and acts as the primary digital employment gateway for vulnerable youth populations. Achieving this extreme security posture requires the platform to implement profound defence in depth architectures, ensuring unparalleled resilience against advanced persistent threats, malicious state sponsored actors, algorithmic manipulation, and complex extraction vectors. The architecture guarantees high assurance security by enforcing strict zero trust network boundaries, utilizing advanced runtime application self-protection mechanisms, and mandating verifiable cryptographic integrity for every single data transaction executed within the application environment.

At its operational core, the platform facilitates a dynamic, high security communication and matchmaking environment for job seekers, private sector employers, and approved external job portals. The system relies on a robust technology stack featuring advanced Artificial Intelligence and Machine Learning frameworks to execute complex vector-based scoring algorithms and Natural Language Processing. By parsing complex data matrices including technical skills, educational backgrounds, user behaviours, and explicit employer requirements, the Artificial Intelligence engine dynamically ranks candidates and recommends opportunities. To maintain this rigorous security posture, all Artificial Intelligence components are strictly compartmentalized within secure execution enclaves, completely preventing any potential data poisoning or algorithmic bias injection while ensuring the Artificial Intelligence operates strictly as a decision support tool rather than an autonomous hiring authority.

Furthermore, the platform integrates an exceptionally secure Application Programming Interface framework to securely ingest job data from third party sites and verify employer identities against official government records via cryptographically signed OAuth tokens. Operating under the absolute highest levels of user safety and digital survivability, the platform protects all permanent data at rest using Advanced Encryption Standard 256 and secures all active network communications with Transport Layer Security version 1.3. By providing a structured, thoroughly verified, and centralized employment portal available comprehensively in both the Arabic and English languages, the system aims to standardize job postings and generate critical employment statistics. Ultimately, this system serves as the premier, highly secure digital infrastructure for the Palestinian workforce, empowering economic growth while guaranteeing absolute data privacy and systemic invulnerability.

# 2. Objectives

The objectives of the Artificial Intelligence Driven Job Matching Platform are meticulously structured into absolute high assurance security goals, critical socioeconomic enablement targets, advanced algorithmic mandates, and secure operational integration directives. These objectives are explicitly designed to guarantee that the platform functions flawlessly as an invulnerable, critical national infrastructure for the Palestinian territory.

- **High Assurance Security Objectives:**
    - Achieve and permanently maintain absolute compliance with stringent Open

Worldwide Application Security Project mandates to safeguard the platform against advanced persistent threats, malicious algorithmic manipulation, and complex data extraction vectors.

  - Implement a rigorous zero trust network architecture across all internal computational boundaries and external Application Programming Interfaces to guarantee absolute data confidentiality for all registered users.
  - Establish a highly resilient disaster recovery operational state guaranteeing digital survivability and continuous availability, specifically mandating exactly a four hour Recovery Time Objective and strictly a one hour Recovery Point Objective during any catastrophic failure scenario.
  - Enforce absolute data protection, extreme privacy, and user safety compliance by utilizing Advanced Encryption Standard 256 for all permanent data at rest within the database and strictly securing all active network communications with Transport Layer Security version 1.3 to completely neutralize packet interception.
  - Prevent all unauthorized employer access by enforcing mandatory authentication via One Time Passwords and executing strict algorithmic verification of employer credentials against official government databases prior to issuing any Verified Employer status.

- **Critical Socioeconomic Infrastructure and Inclusivity Objectives:**
  - Operate flawlessly as the premier critical digital infrastructure for the Palestinian workforce, specifically empowering the economic advancement of vulnerable youth populations between fifteen and twenty-nine years old, including young women and marginalized community members.
  - Ensure a completely safe, equitable, and highly accessible digital environment by strictly adhering to Web Content Accessibility Guidelines version 2.1 Level AA standards, explicitly guaranteeing that individuals with disabilities can securely navigate the platform using specialized assistive technologies.
  - Standardize the historically fragmented national labour market by creating a unified, mathematically consistent job posting schema utilizing the Schema dot org

JobPosting standard to guarantee absolute semantic compatibility across all internal listings and external integrated portals.

- Provide a flawless multilingual user experience by supporting both the Arabic and English languages comprehensively across all graphical user interfaces and Natural Language Processing models, including full Right To Left text direction support.

- **Advanced Artificial Intelligence and Algorithmic Boundaries Objectives:**
  - Deploy a highly sophisticated, secure, and isolated vector-based scoring algorithm to precisely calculate match percentages between job seekers and active postings based on complex data matrices including skill overlap, educational background, required training, and physical location parameters.
  - Utilize advanced Natural Language Processing to accurately parse unstructured resume documents and perfectly understand the deep semantic meaning behind job descriptions, deliberately moving the platform completely beyond legacy keyword matching limitations.
  - Ensure absolute algorithmic fairness and strictly prevent automated hiring bias by physically confining the Artificial Intelligence components strictly to decision support, data extraction, and candidate ranking functions. The Artificial Intelligence is explicitly prohibited from making final employment decisions, executing automated candidate rejections, or possessing autonomous hiring authority.
  - Provide an extremely robust two-way recommendation engine that simultaneously pushes highly ranked job opportunities to active youth seekers while generating refined, ranked shortlists of the absolute highest matching candidates for verified private sector employers.

- **Operational Integration and Labor Market Analytics Objectives:**
  - Facilitate secure, real time data synchronization with approved external third-party job portals, such as jobs dot ps, through a highly secure RESTful Push Application Programming Interface framework utilizing strict OAuth 2.0 cryptographic tokens.

    - Execute highly secure data exchanges with critical Palestinian government databases to continuously authenticate employer registration credentials and rigorously validate job seeker educational certificates without ever exposing the core database ledger.
    - Generate comprehensive, real time employment statistics, geographic distribution reports, and critical skill demand analytics to constantly empower the Ministry of Labor and the Palestinian Employment Fund with highly actionable labour market intelligence.
    - Deploy a centralized, high performance notification engine capable of securely dispatching customizable email templates, internal application alerts, and critical Short Message Service updates through approved telecommunication gateways to keep all users continuously informed regarding their employment journey.

# 3. Scope

**In Scope:** The following modules, architectural components, and functionalities are strictly within the scope of the AI-Driven Job Matching Platform project:

- **User Management and Verification:**
    - Complete user registration and profile management workflows for job seekers, allowing multi-level onboarding to capture education, work experience, skills, and personal bios.
    - Employer registration workflows requiring official company identification numbers, combined with mandatory authentication via One-Time Passwords (OTP) and official government database verification to issue "Verified Employer" badges.
    - Administrative user management capabilities allowing Ministry of Labor officials to approve, ban, or deactivate user accounts, manage static reference taxonomies (skills, occupations), and maintain complete audit logs of all platform actions.
- **Job Posting and Search Engine:**
    - Structured job creation tools allowing employers to define job summaries, contract

types, required education levels, physical or hybrid work formats, and specific application deadlines.

  - An advanced search engine interface for job seekers supporting both basic and semantic search modes, utilizing complex filters such as salary ranges, required languages, and industry sectors.

- **Artificial Intelligence and Matching Components (Explicit Boundaries):**
  - **In Scope for AI:** The utilization of AI to perform automated resume parsing (extracting structured data from PDFs or DOCX files), executing NLP to identify semantic relationships in text, and calculating vector-based match scores to rank candidates and jobs.
  - **In Scope for AI:** The generation of personalized job recommendations based on collaborative filtering (similar user behaviours) and content-based filtering (user interests and preferences).
  - **AI Boundary Definition:** The AI components are strictly limited to data extraction, semantic scoring, and recommendation generation. The AI is explicitly constrained from making final employment decisions, automatically rejecting candidates without employer review, or executing any automated hiring actions. It functions purely as a decision-support and ranking mechanism.

- **Integration and API Framework:**
  - The deployment of RESTful APIs utilizing JSON data formats and OAuth 2.0 authentication to enable both pull and push synchronization models with approved external job sites.
  - Secure data exchanges with government databases to verify employer registration credentials and validate job seeker educational certificates.

- **System Telemetry, Analytics, and Notifications:**
  - Comprehensive monitoring of user activities, application rates, and system performance metrics, visualized through custom report builders and administrative dashboards.

    - A centralized notification system capable of dispatching customizable email templates, in-app alerts, and critical Short Message Service (SMS) updates through third-party gateways.
- **Localization and Cultural Adaptation:**
    - Full multilingual user interface support across the entire platform in both the Arabic and English languages, including right-to-left text direction support and language-specific NLP models.

**Out of Scope:** The following items, operational boundaries, and features are explicitly excluded from this phase of the project:

- **Comprehensive Labor Market Information System (LMIS):**
    - The development and implementation of the complete, overarching LMIS architecture. This current phase is strictly limited to the core Job Matching Platform component.
- **Hardware and Physical Infrastructure Procurement:**
    - The physical purchasing, provisioning, or installation of the server hardware, data center racks, or load balancing devices required for hosting the solution.
- **External Integrations Beyond Specified Entities:**
    - Any integration, API development, or data synchronization with systems, global job boards, or third-party databases not explicitly approved by the Ministry of Labor and Palestinian Employment Fund in this document.
- **End-User Training and Long-Term Maintenance:**
    - The execution of massive training programs for general public end-users, as well as multi-year technical support, help desk operations, and long-term maintenance contracts, which will be addressed in separate future agreements.
- **Native Mobile Application Engineering:**
    - The coding, deployment, or publishing of dedicated native applications for iOS or

Android operating systems. The platform relies entirely on a responsive web-based architecture for mobile accessibility during this phase.

# 4. Functional Requirements

The functional requirements of the Artificial Intelligence Driven Job Matching Platform strictly define the secure, high assurance operational capabilities of the core matchmaking engine. Operating as a critical national socioeconomic infrastructure, the platform mandates flawless execution of complex algorithmic workflows, encompassing secure multi-level user onboarding, highly accurate resume parsing via Natural Language Processing, and dynamic vector based scoring. Because the system facilitates employment for vulnerable youth populations, all functional interactions are strictly compartmentalized to ensure absolute algorithmic fairness and privacy. The Artificial Intelligence components are explicitly constrained to decision support and ranking functionalities, empowering job seekers with personalized, mathematically precise recommendations while simultaneously providing verified employers with ranked candidate shortlists. Every micro interaction, from calculating statistical confidence scores during data extraction to processing employer search queries, is engineered to provide a seamless, highly responsive, and totally verifiable user experience within a strictly controlled digital environment.

- **FR-001:** The system shall implement an Artificial Intelligence driven matching algorithm to seamlessly connect registered job seekers with highly relevant employment opportunities.

- **FR-002:** The matching engine shall utilize a sophisticated vector based scoring algorithm to calculate a precise mathematical match percentage between job seeker profiles and active job postings.

- **FR-003:** The scoring algorithm shall explicitly calculate match percentages based on a complex data matrix including skill overlap, educational background, required training, physical location proximity, experience range, and salary expectations.

- **FR-004:** The system shall automatically generate customized shortlists containing up to exactly one hundred top matching candidates for every single active job posting to explicitly assist employers during their selection process.

- **FR-005:** The Artificial Intelligence engine shall utilize advanced Natural Language

Processing to accurately comprehend the deep semantic meaning embedded within job descriptions and uploaded resumes, deliberately moving completely beyond basic legacy keyword matching.

- **FR-006:** The Natural Language Processing engine shall simultaneously execute keyword analysis and semantic analysis on all posted job descriptions to systematically identify key attributes like required skills, experience levels, and specific job categories.
- **FR-007:** The platform shall provide authorized administrators with a configuration interface to manually define and adjust a minimum match threshold percentage, such as sixty percent, which strictly dictates which job matches are visibly displayed to end users.
- **FR-008:** The platform shall dynamically rank and visually order all displayed job postings for each individual job seeker strictly based on their calculated individual match percentage, ensuring the absolute most relevant opportunities appear first.
- **FR-009:** The Artificial Intelligence engine shall perform bi directional matching, ensuring job seekers receive ranked job opportunities while verified employers simultaneously receive reverse matches containing ranked lists of highly suitable job seekers.
- **FR-010:** The platform shall expose a configuration panel allowing system administrators to manually adjust the mathematical importance and weighted parameters of different matching factors.
- **FR-011:** The system shall execute Artificial Intelligence powered resume parsing to intelligently extract highly structured information from raw uploaded document files such as PDF and DOCX formats.
- **FR-012:** The resume parsing engine shall accurately identify and extract discrete data points including personal details, contact information, education history, work experience, skills, certifications, and professional achievements.
- **FR-013:** The natural language parsing engine must maintain comprehensive multilingual support, possessing the explicit capability to seamlessly parse resumes written in both the Arabic and English languages.
- **FR-014:** The parsing module shall intelligently identify unstructured skills mentioned within

a resume and automatically standardize them against the centralized platform skill taxonomy to facilitate accurate vector matching.

- **FR-015:** The Artificial Intelligence engine shall calculate and explicitly append statistical confidence scores to every single piece of extracted information during the automated resume parsing workflow.
- **FR-016:** The graphical user interface shall visibly highlight any parsed data fields possessing low confidence scores to explicitly prompt the job seeker for mandatory manual verification.
- **FR-017:** The system shall present an interactive review screen allowing job seekers to manually edit, correct, and save any erroneously parsed resume information prior to finalizing their profile construction.
- **FR-018:** The platform shall power a personalized job recommendation engine that comprehensively evaluates the job seeker profile, explicit user preferences, and historical platform activity to proactively push suitable job opportunities.
- **FR-019:** The recommendation engine shall simultaneously recommend highly qualified job seekers directly to verified employers based on specific job requirements and identical vector matching criteria.
- **FR-020:** The system shall implement collaborative filtering algorithms to intelligently recommend employment opportunities based on the historical activities and similar behaviors of comparable user profiles.
- **FR-021:** The system shall implement content-based filtering algorithms to proactively recommend new job postings that share profound semantic similarities with postings the user previously engaged with.
- **FR-022:** The personalized recommendation algorithm shall explicitly factor in customized user preferences regarding physical location, specific salary expectations, and desired work arrangements.
- **FR-023:** The system shall dynamically generate candidate recommendation dashboards for verified employers, ranking potential candidates based strictly on absolute match

quality.

- **FR-024:** The candidate recommendation interface shall visually highlight the specific matched strengths of each candidate while simultaneously identifying potential skill gaps compared directly to the job description.

- **FR-025:** The platform shall allow verified employers to establish strict minimum qualification thresholds to facilitate the automatic filtering and immediate exclusion of unqualified candidates from their recommendation dashboards.

- **FR-026:** The system shall provide employers with an advanced search interface to manually query the recommended candidate database utilizing highly granular filtering options.

- **FR-027:** The candidate recommendation engine shall strictly respect and legally enforce all job seeker privacy settings, ensuring private profiles are completely excluded from employer recommendation visibility.

- **FR-028:** The recommendation interface shall provide employers with predictive insights regarding candidate market availability, accurate salary expectations, and overall organizational potential fit.

- **FR-029:** The system shall present a user interface control allowing verified employers to securely save promising recommended candidates directly into customized talent pools for future recruitment tracking.

# 5. Non-Functional Requirements-

The non-functional requirements establish the absolute high assurance security posture, rigorous performance thresholds, and critical operational constraints necessary to sustain this stringent high assurance infrastructure. Given the extreme sensitivity of the processed data and the platform's role in national labour logistics, the architecture demands a zero-trust security model across all algorithmic boundaries. To definitively neutralize advanced persistent threats and algorithmic manipulation, the system mandates strict allow listing for all input validation, completely parameterized database queries, and robust semantic sanitization to defeat sophisticated Artificial Intelligence specific attack vectors like prompt injection and Retrieval

Augmented Generation context poisoning. Furthermore, the platform guarantees absolute data confidentiality and digital survivability by encrypting all permanent data at rest with Advanced Encryption Standard 256, securing all active network transits with Transport Layer Security version 1.3, and employing continuous runtime application self-protection. Beyond security, the processing infrastructure is highly optimized to flawlessly execute massive vector scoring calculations for thousands of concurrent users without any architectural degradation, ensuring the platform remains highly available, responsive, and invulnerable under maximum operational loads.

- **NFR-001:** To satisfy stringent high assurance security mandates, the system architecture must strictly mandate the utilization of parameterized queries for absolutely all database interactions to permanently eliminate any possibility of Structured Query Language injection vulnerabilities.

- **NFR-002:** The system architecture must enforce a rigorous zero trust input validation mechanism, explicitly requiring strict allow listing for all incoming data payloads, user profile submissions, and parsed resume text before any backend processing occurs.

- **NFR-003:** The Artificial Intelligence components and Natural Language Processing engines must be fortified with comprehensive defence mechanisms specifically engineered to neutralize advanced Artificial Intelligence specific vectors, explicitly including robust semantic sanitization against malicious prompt injection attacks.

- **NFR-004:** The vector-based scoring architecture must implement strict contextual boundaries to definitively prevent Retrieval Augmented Generation context poisoning, ensuring that malicious adversarial inputs hidden within uploaded resumes cannot manipulate the semantic matching models or influence broader platform recommendations.

- **NFR-005:** The application must deploy advanced runtime application self-protection tools to provide continuous integrity monitoring across all Artificial Intelligence execution enclaves and core algorithmic files, immediately flagging any anomalous code modifications.

- **NFR-006:** All permanent data at rest, explicitly including the high dimensional vector embeddings utilized by the Artificial Intelligence engine for matching calculations, must be meticulously encrypted using Advanced Encryption Standard 256 cryptographic protocols.

- **NFR-007:** The platform must protect all sensitive matching matrices and active network communications by strictly enforcing Transport Layer Security version 1 point 3 across all internal microservices and external system boundaries.
- **NFR-008:** The Artificial Intelligence matching engine must reliably complete sophisticated mathematical vector calculations for individual job and candidate matches in strictly less than five seconds to guarantee a highly responsive user experience.
- **NFR-009:** The system must flawlessly execute massive computational batch operations, such as bulk candidate matching and algorithmic model updates, within a timeframe strictly proportional to the batch size while absolutely not exceedingly exactly two minutes for standard operational loads.
- **NFR-010:** The Artificial Intelligence algorithms and underlying processing infrastructure must be highly optimized to seamlessly sustain simultaneous vector scoring and resume parsing calculations for a minimum of one thousand concurrent users without experiencing any architectural performance degradation.

# 6. System Architecture

**Architecture Type:** High-Assurance Zero-Trust Microservices Architecture

**Description:** The Artificial Intelligence Driven Job Matching Platform utilizes a highly robust, high-assurance microservices architecture fundamentally engineered around strict Zero-Trust Network Architecture principles. Because this platform operates as a critical national socioeconomic infrastructure for the Palestinian territory, the default systemic posture explicitly assumes that the network is constantly hostile. Consequently, there is absolutely no implicit trust granted to any user, internal microservice, or external Application Programming Interface request based solely on their network location. Every single interaction crossing a boundary must be explicitly authenticated, cryptographically authorized, and continuously validated before any data processing is permitted. The entire environment is meticulously segmented into strictly isolated subnets, placing the public-facing Web Application Firewalls and load balancers in a secure demilitarized zone, while the core database ledgers and Artificial Intelligence execution enclaves are buried deep within highly restricted, private subnets completely severed from direct internet routing.

At the core of this highly secure ecosystem resides the Artificial Intelligence Matching Engine. To guarantee the absolute safety of vulnerable user profiles and prevent malicious algorithmic manipulation, the Artificial Intelligence components (including the Curriculum Vitae Parser, Semantic Matching algorithms, and Recommender Systems) are physically and logically quarantined within secure execution enclaves. When a job seeker uploads a resume or an employer submits a job posting, the unstructured data first passes through a dedicated Data Sanitization Agent. This agent meticulously strips all Personally Identifiable Information and sanitizes the input to explicitly neutralize advanced Artificial Intelligence specific attacks like prompt injection and Retrieval Augmented Generation context poisoning. Only after absolute sanitization is the data securely forwarded to the Natural Language Processing models for semantic analysis and high dimensional vector embedding generation. This strict compartmentalization ensures the algorithmic models operate entirely on anonymized matrices, utterly preventing data leakage.

Furthermore, the overarching system architecture acts as the central nucleus of the broader Labor Market Information System. The platform seamlessly and securely orchestrates data exchanges with external entities, including the Ministry of Labor databases, the Palestinian Employment Fund databases, recognized educational institutions, and approved third-party job sites like jobs dot ps. All external integrations are executed through a rigidly controlled Application Programming Interface gateway utilizing strict OAuth 2.0 cryptographic tokens and JSON data formats. By combining highly isolated AI enclaves, continuous runtime integrity monitoring, and absolute Zero-Trust verification methodologies, the system guarantees an invulnerable infrastructure capable of securely evaluating millions of vector coordinates while perfectly protecting the Palestinian workforce.

### Core System Interactions and Secure AI Processing Sequence Diagram:

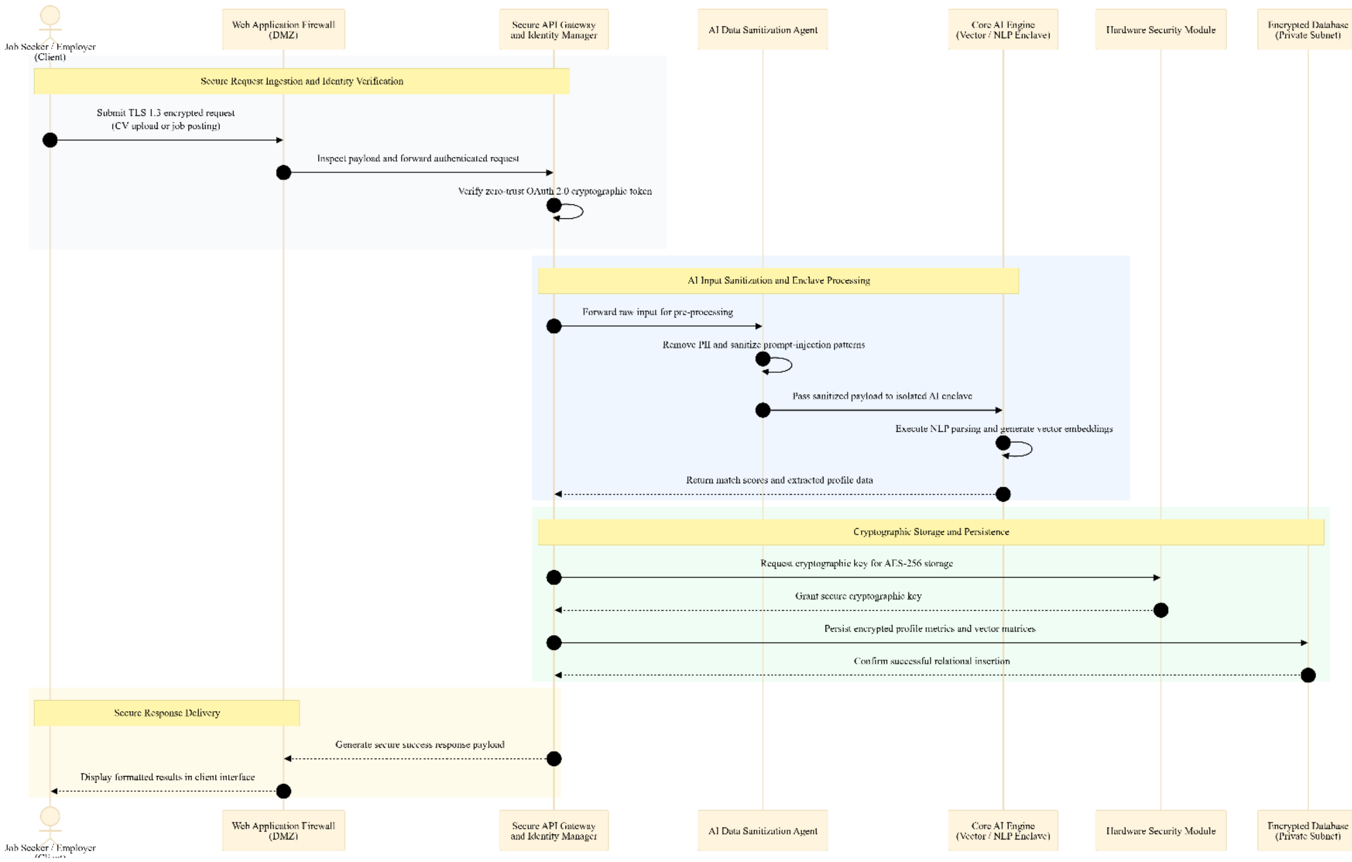


*Figure 1 Secure AI-Based Recruitment Processing Sequence Diagram*

# 7. Technical Stack

The technical stack of the Artificial Intelligence Driven Job Matching Platform is meticulously selected to guarantee stringent high assurance security compliance, extreme computational performance, and impenetrable data security. The presentation layer utilizes responsive web design methodologies to ensure flawless execution across desktop computers, laptops, tablets, and mobile smartphones. Because the platform manages complex natural language translation between Arabic and English, the client-side interfaces rely on highly optimized frameworks capable of seamlessly rendering right to left text directions without latency.

To satisfy the extreme security mandates of this critical infrastructure, the backend processing environments integrate enterprise-grade security software explicitly designed to protect sensitive user profiles and job listings. The architecture mandates the integration of Hardware Security Modules to physically protect the lifecycle of all critical cryptographic keys, preventing even administrative personnel from extracting the master encryption passwords. Furthermore, the stack strictly implements Federal Information Processing Standards (FIPS 140-2 and FIPS 140-3) validated cryptographic libraries, ensuring that all AES-256 data at rest and Transport Layer Security 1.3 data in transit are secured using mathematically provable, government-grade encryption methodologies. This sophisticated combination of modern Artificial Intelligence frameworks and absolute high-assurance security tools guarantees an unprecedented level of digital survivability.

| **Layer** | **Technology** |
|---|---|
| **Frontend User Interface (GUI)** | HTML5, CSS3, Responsive JavaScript Frameworks (Enterprise Standard) |
| **Backend Business Logic** | High-Performance Server-Side Language configured for secure runtime |
| **Database Management System** | Enterprise Relational Database with high performance and scalability |
| **Artificial Intelligence Framework** | Advanced AI and Machine Learning Frameworks (Isolated Enclave) |
| **Cryptographic Modules** | FIPS 140-2 and FIPS 140-3 Validated Cryptographic Libraries |
| **Hardware Key Management** | Dedicated Hardware Security Modules (HSM) |
| **Data Encryption Standard (Rest)** | Advanced Encryption Standard 256-bit (AES-256) |
| **Data Encryption Standard (Transit)** | Transport Layer Security (TLS) Version 1.3 |
| **Authentication & Authorization** | OAuth 2.0, OpenID Connect, Multi-Factor Authentication Tokens |
| **API Integration Protocol** | RESTful Push API utilizing JSON data formats |
| **Network Security Infrastructure** | Advanced Web Application Firewalls (WAF) and Load Balancers |

| | |
|---|---|
| **Hosting Environment** | Scalable High-Availability Cloud, On-Premises, or Hybrid Infrastructure |
| **Client Environments Supported** | Chrome, Firefox, Safari, Edge, iOS Browsers, Android Browsers |

# 8. Interfaces & Integrations

The Application Programming Interface architecture for the Artificial Intelligence Driven Job Matching Platform is meticulously engineered to establish an impenetrable digital perimeter capable of sustaining critical national socioeconomic logistics. Because the system directly integrates with highly sensitive Palestinian Ministry of Labor databases and processes the Personally Identifiable Information of vulnerable youth populations, absolute high assurance security is mandatory across every single computational boundary. To strictly comply with these rigorous security mandates, the architecture explicitly dictates that absolutely every internal and external Application Programming Interface endpoint requires mutual Transport Layer Security 1.3 cryptographic tunnelling. Furthermore, all requests must be authenticated using cryptographically signed JSON Web Tokens featuring strict audience validation, ensuring that a token issued for a specific microservice cannot be maliciously reused elsewhere. To definitively prevent distributed denial of service attacks, robotic spamming, and credential brute forcing, the entire integration framework is shielded by aggressive rate limiting algorithms deployed at the Web Application Firewall layer.

The platform relies on a vast array of specialized interfaces to orchestrate real time job synchronization, government verification, and Artificial Intelligence computations. By mandating JSON data formats and strict OAuth 2.0 authorization, the system guarantees seamless yet totally secure interoperability between internal processing enclaves and approved external third-party platforms.

- **Internal Endpoint POST /api/v1/auth/authenticate:** This highly secure endpoint handles the initial user login request, validating the submitted credentials and triggering the mandatory phishing resistant FIDO2 or WebAuthn multi factor authentication flow before issuing a cryptographically signed JSON Web Token.
- **Internal Endpoint POST /api/v1/profile/resume/parse:** This critical internal interface

securely routes uploaded portable document format or Word document resumes directly into the isolated Artificial Intelligence execution enclave, initiating the Natural Language Processing pipeline to accurately extract highly structured educational and skill metrics.

- **Internal Endpoint GET /api/v1/matching/candidates/score:** This vector calculation endpoint receives a specific job posting identifier and aggressively queries the Artificial Intelligence matching engine to mathematically calculate and return a ranked shortlist of up to exactly one hundred top matching job seekers based on complex skill overlap and semantic matrices.
- **Internal Endpoint POST /api/v1/jobs/search/semantic:** This advanced internal search interface processes unstructured queries submitted by job seekers, utilizing deep semantic understanding rather than legacy keyword matching to return highly relevant employment opportunities rendered in strictly less than two seconds.
- **External Integration External Job Portals Push API:** This secure RESTful integration allows approved external third-party platforms, such as jobs dot ps, to submit and continuously synchronize job opportunities directly into the central matching system, requiring explicit Application Programming Interface keys, IP whitelisting, and strict payload validation.
- **External Integration Ministry of Labor Verification Gateway:** This critical government interface establishes a highly restricted outbound connection to official Palestinian databases to automatically cross reference and rigorously verify employer registration numbers and corporate identification metrics before granting the Verified Employer badge.
- **External Integration Educational Credentials Validation API:** This external data enrichment interface allows the platform to securely connect with recognized Palestinian educational institution databases to mathematically verify the authenticity of degrees and training certificates claimed by youth job seekers.
- **External Integration Short Message Service Dispatch Gateway:** This external communication interface connects strictly to approved telecommunication providers via secure protocols to rapidly dispatch critical time sensitive alerts, one-time passwords, and account activation codes directly to user mobile devices.
- **External Integration Secure SMTP Email Dispatcher:** This external notification

interface connects to enterprise email service providers to securely transmit customizable, dynamically generated email templates encompassing registration confirmations, weekly job recommendations, and platform updates.

# 9. Data Model

The Data Model for the Artificial Intelligence Driven Job Matching Platform is architected to operate under the absolute highest echelons of data security, privacy protection, and relational integrity. Because the system serves as critical socioeconomic infrastructure processing highly sensitive profiles for vulnerable youth populations, the database schema physically and logically segregates Personally Identifiable Information from standard operational metrics. To guarantee absolute compliance with stringent high assurance security mandates, all data at rest is strictly secured using AES-256-GCM managed via HSM. This rigorous cryptographic posture ensures that even in the event of a catastrophic physical server breach, sensitive elements such as mobile numbers, national identification parameters, and granular personal biographies remain mathematically impenetrable.

Special attention is explicitly paid to the structuring of geographic location data and individual user identities. To facilitate precise vector-based proximity scoring without ever exposing exact residential coordinates to unauthorized entities, physical addresses are systematically normalized into tiered geospatial fields. Current addresses and permanent residences are securely stored utilizing specialized spatial data types. This sophisticated geographic structuring allows the Artificial Intelligence engine to accurately calculate commute distances and regional match percentages while strictly preserving job seeker anonymity prior to any mutual employer agreement. Furthermore, all unstructured data extracted during the automated resume parsing workflows is immediately structured into strictly typed relational tables, ensuring the natural language algorithms evaluate highly sanitized data vectors rather than raw, potentially toxic input documents.

**Entities and Key Fields:**

| **Entity** | **Field** | **Type** |
|---|---|---|
| JobSeeker | id | Universally Unique Identifier (Primary Key) |

| | | |
|---|---|---|
| JobSeeker | first_name | String (AES-256-GCM Encrypted PII) |
| JobSeeker | last_name | String (AES-256-GCM Encrypted PII) |
| JobSeeker | email_address | String (Unique Constraint, AES-256-GCM Encrypted) |
| JobSeeker | mobile_number | String (Unique Constraint, AES-256-GCM Encrypted) |
| JobSeeker | password_hash | String (Argon2id Cryptographic Hash) |
| JobSeeker | gender | Categorical Enumeration |
| JobSeeker | current_address | Geospatial Point (Latitude and Longitude, Encrypted) |
| JobSeeker | permanent_residence | Geospatial Point (Latitude and Longitude, Encrypted) |
| JobSeeker | recent_salary_expectation | Numeric Decimal (Optional Field) |
| JobSeeker | social_media_links | JSON Array (LinkedIn, X profiles) |
| JobSeeker | personal_statement_bio | Large Text Object (Sanitized for Prompt Injection) |
| JobSeeker | account_activation_status | Boolean (Verified via SMS One Time Password) |
| SeekerEducation | id | Universally Unique Identifier (Primary Key) |
| SeekerEducation | job_seeker_id | Universally Unique Identifier (Foreign Key) |
| SeekerEducation | degree_level | Standardized Enumeration Taxonomy |
| SeekerEducation | institution_name | String |

| | | |
|---|---|---|
| SeekerExperience | id | Universally Unique Identifier (Primary Key) |
| SeekerExperience | job_seeker_id | Universally Unique Identifier (Foreign Key) |
| SeekerExperience | job_title | String (Standardized to Occupations Catalog) |
| SeekerExperience | duration_months | Integer |
| SeekerSkill | id | Universally Unique Identifier (Primary Key) |
| SeekerSkill | job_seeker_id | Universally Unique Identifier (Foreign Key) |
| SeekerSkill | skill_taxonomy_id | Universally Unique Identifier (Standardized Reference) |
| SeekerSkill | confidence_score | Decimal (Generated by AI Resume Parser) |
| Employer | id | Universally Unique Identifier (Primary Key) |
| Employer | company_name | String |
| Employer | primary_email | String (Unique Constraint, Encrypted) |
| Employer | mobile_number | String (Unique Constraint, Encrypted) |
| Employer | company_national_id | String (AES-256-GCM Encrypted, Government Verified) |
| Employer | registration_number | String (AES-256-GCM Encrypted) |
| Employer | verification_badge | Boolean (Approved via Ministry of Labor Database) |
| Employer | industry_sector | Standardized Enumeration Taxonomy |

| | | |
|---|---|---|
| Employer | company_size | Categorical Enumeration |
| Employer | physical_address | Geospatial Point (Latitude and Longitude) |
| Employer | company_description | Large Text Object |
| Employer | supplementary_documents | Blob Array (Securely Stored Registration Files) |
| JobPosting | id | Universally Unique Identifier (Primary Key) |
| JobPosting | employer_id | Universally Unique Identifier (Foreign Key) |
| JobPosting | job_title | String |
| JobPosting | structured_summary | Large Text Object (Semantic Search Indexed) |
| JobPosting | required_skills | JSON Array (Mapped to Skill Taxonomy) |
| JobPosting | contract_type | Categorical Enumeration |
| JobPosting | required_education | Standardized Enumeration Taxonomy |
| JobPosting | work_format | Enumeration (Physical, Hybrid, Remote) |
| JobPosting | application_deadline | Timestamp |
| JobPosting | source_platform_tag | String (Identifies External Third Party Sites) |
| JobPosting | unique_external_job_id | String (For API Synchronization Tracking) |
| AIVectorEmbedding | id | Universally Unique Identifier (Primary Key) |
| AIVectorEmbedding | reference_id | Universally Unique Identifier (Links to Seeker or Job) |

| | | |
|---|---|---|
| AIVectorEmbedding | semantic_matrix | High Dimensional Float Array (AES-256-GCM Encrypted) |
| AuditLog | id | Universally Unique Identifier (Primary Key) |
| AuditLog | user_id | Universally Unique Identifier (Actor Identifier) |
| AuditLog | administrative_action | String (Detailed Description of the Event) |
| AuditLog | timestamp | Timestamp (Immutable) |

## 10. Configuration Parameters

The operational environment of the Artificial Intelligence Driven Job Matching Platform relies upon a highly exhaustive matrix of configuration parameters, strict physical system limits, application timeouts, and heavily protected environment variables. To successfully fulfil the stringent performance requirements and scalability mandates outlined by the Palestinian Employment Fund, these values define the exact computational boundaries of the sixty-four-bit architecture. The system must flawlessly sustain massive concurrent traffic spikes without exceeding strict central processing unit or random-access memory thresholds. Furthermore, to protect the Natural Language Processing engines from computational exhaustion or malicious resource consumption, the architecture explicitly defines rigid Artificial Intelligence token usage limits for every single interaction. To guarantee total cryptographic security, these environment variables are explicitly prohibited from being hardcoded into the software repository, and are instead dynamically injected into the runtime environment utilizing an isolated enterprise secrets vault.

| Parameter | Value | Description |
|---|---|---|
| PAGE_LOAD_TIMEOUT_SEC | 3 | The maximum allowable second duration for rendering any standard system page under normal load conditions to guarantee a highly |

| | | |
|---|---|---|
| | | responsive user interface. |
| SEARCH_RESULT_TIMEOUT_SEC | 2 | The strictly enforced maximum threshold in seconds allowed for the database to return standard search queries to the graphical interface. |
| AI_MATCHING_TIMEOUT_SEC | 5 | The maximum allowable processing time for the Artificial Intelligence engine to execute complex multi-dimensional vector calculations for individual candidate matches. |
| BATCH_PROCESSING_TIMEOUT_MIN | 2 | The absolute maximum duration in minutes permitted for executing massive computational operations such as bulk platform wide candidate scoring. |
| MAX_CONCURRENT_USERS_NORMAL | 1000 | The baseline architectural requirement dictating the number of users who must be perfectly supported simultaneously during standard operational hours. |
| MAX_CONCURRENT_USERS_PEAK | 5000 | The extreme upper limit of simultaneous platform connections the infrastructure must mathematically support during periods of unprecedented traffic. |
| MAX_APPLICATIONS_PER_MINUTE | 100 | The aggressive rate limiting threshold defining the maximum volume of job applications the central server will process globally every sixty seconds. |
| DAILY_NEW_POSTING_CAPACITY | 500 | The baseline expected volume of structurally formatted new job opportunities seamlessly ingested into the central database ledger every single day. |
| DAILY_NEW_REGISTRATION_CAP | 1000 | The baseline volume of new youth job seeker and employer profiles the platform is mathematically engineered |

| | | |
|---|---|---|
| | | to authenticate and provision daily. |
| CPU_UTILIZATION_THRESHOLD | 80 Percent | The absolute maximum central processing unit usage allowed before the high availability load balancers automatically trigger horizontal scaling protocols. |
| MEMORY_UTILIZATION_THRESHOLD | 80 Percent | The strict ceiling for random access memory consumption enforced across all active application servers to prevent catastrophic memory overflow errors. |
| INITIAL_STORAGE_ALLOCATION_TB | 5 | The massive physical disk space specifically provisioned to securely house encrypted user profiles, parsed resume files, and immutable audit logs for the first year. |
| MINIMUM_SEEKER_CAPACITY | 100000 | The mandatory baseline storage volume guaranteeing flawless performance for at least one hundred thousand highly detailed registered youth profiles. |
| MINIMUM_EMPLOYER_CAPACITY | 10000 | The mandatory baseline storage volume explicitly ensuring the secure persistence of at least ten thousand registered corporate entities and their verification documents. |
| ACTIVE_JOB_POSTING_CAPACITY | 50000 | The mandatory baseline database capacity explicitly engineered to manage and index fifty thousand simultaneous, searchable job listings without search degradation. |
| AI_MINIMUM_MATCH_THRESHOLD | 60 Percent | The dynamically adjustable mathematical baseline determining exactly which generated job recommendations are visibly surfaced to the end user dashboard. |
| MAX_SHORTLIST_CANDIDATES | 100 | The strict numerical cap defining the |

| | | |
|---|---|---|
| | | exact maximum amount of top matching applicant profiles pushed to a verified employer for any single open vacancy. |
| AI_DAILY_TOKEN_LIMIT_SEEKER | 25000 | The explicit computational ceiling of Artificial Intelligence semantic processing tokens allocated per job seeker daily to completely prevent denial of wallet attacks. |
| AI_DAILY_TOKEN_LIMIT_EMPLOYER | 150000 | The elevated computational ceiling of Artificial Intelligence tokens granted to verified employers to facilitate bulk candidate matching and complex semantic searches. |
| PROD_AIJMP_DB_HOST | Injected String | The highly restricted, cryptographically vaulted environment variable defining the precise internal network coordinates of the air gapped primary database ledger. |
| PROD_AIJMP_JWT_AUDIENCE | Injected String | The crucial security variable explicitly defining the authorized audience parameter strictly required to successfully validate all incoming cryptographically signed web tokens. |
| PROD_AIJMP_MOL_API_KEY | Injected String | The extremely sensitive, automatically rotated access token required to establish a mutual Transport Layer Security connection with the Ministry of Labor verification database. |

# 11. User Access & Roles

The Artificial Intelligence Driven Job Matching Platform employs a sophisticated, defence in depth authorization architecture blending rigid Role Based Access Control with dynamic Attribute Based Access Control paradigms. This complex amalgamation guarantees that users operate exclusively within their strictly permitted digital boundaries while allowing the platform to evaluate real time contextual variables before granting access to highly sensitive socioeconomic data. The underlying Attribute Based Access Control logic allows the system to dynamically evaluate specific user attributes, geographic locations, and verification statuses. For instance, the system explicitly evaluates whether a corporate entity possesses a valid verification badge issued by the Ministry of Labor before permitting that entity to view the Artificial Intelligence generated candidate shortlists or extract sensitive applicant profiles. By combining rigid roles with fluid attribute validation, the architecture inherently protects vulnerable youth populations and perfectly neutralizes unauthorized privilege escalation attempts.

Crucially, to establish absolute security compliance across this critical national infrastructure, the system explicitly dictates that all system access requires mandatory, phishing resistant Multi Factor Authentication prior to being granted any internal access. While legacy platforms may rely on standard text messages, this high assurance environment strictly mandates the utilization of FIDO2 or WebAuthn cryptographic protocols for elevated roles, supported by secure One Time Passwords for standard users. This rigorous security posture means that submitting a valid username and encrypted password combination is merely the initial step. Users must subsequently verify their physical identity through advanced hardware security keys, biometric authenticators, or secure mobile channels, ensuring that compromised primary credentials never lead to unauthorized account breaches or data exfiltration.

The system handles user access by defining specific roles and their associated permissions.

The **System Administrator (Ministry of Labor Admin and Management)** role is granted an access level of **Elevated Administrative Privileges**. Users assigned to this role must authenticate via **Username, Password, and Mandatory Phishing Resistant Multi Factor Authentication utilizing FIDO2 or WebAuthn protocols**. This role is primarily responsible for: comprehensively managing the platform configuration, viewing and modifying static reference taxonomies such as the occupations catalog and skill lists, and executing high level content management including publishing news, updates, and Help Center articles. Administrators are critically responsible for user governance; they possess the explicit authority to view, approve, ban, or deactivate user accounts, reset account credentials, and monitor global account statuses to maintain absolute community safety. Furthermore, they manage the integration logs for external third-party portals and hold the authority to manually verify employer registrations if the automatic government database verification fails. To ensure total accountability, every single action executed by a System Administrator is permanently recorded within an immutable security audit log.

The **Government Official (Platform Management via Ministry of Labor and Palestinian Employment Fund)** role is granted an access level of **Advanced Analytical Access**. Users assigned to this role must authenticate via **Username, Password, and Mandatory Phishing Resistant Multi Factor Authentication utilizing FIDO2 or WebAuthn protocols**. This role is primarily responsible for: securely accessing labour market insights, generating real time employment statistics, reviewing geographic distribution reports for jobs and candidates, and analyzing skill demand trends to identify emerging national requirements. Officials operating under this role can utilize custom report generation frameworks to extract critical performance metrics and employment outcomes. However, to maintain strict privacy boundaries, this role is mathematically restricted from directly modifying individual user records, manipulating active job postings, or bypassing data redaction protocols when viewing personally identifiable information.

The **Verified Employer** role is granted an access level of **Standard Authenticated Organization**. Users assigned to this role must authenticate via **Username, Password, and Mandatory Multi Factor Authentication utilizing One Time Passwords**. This role is primarily responsible for: executing comprehensive profile management through a strict two-level registration workflow. Level one captures the company name, email, mobile number, company ID, and registration number. Level two captures the website, industry, company size, physical address, company description, and uploaded supplementary registration documents. Following successful automatic or manual verification, this role can structurally define and publish new job opportunities, configure application deadlines, manually search the job seeker database using advanced filtering, and review highly customized candidate shortlists generated directly by the Artificial Intelligence matching algorithm.

The **Registered Job Seeker** role is granted an access level of **Standard Authenticated Candidate**. Users assigned to this role must authenticate via **Username, Password, and Mandatory Multi Factor Authentication utilizing One Time Passwords**. This role is primarily responsible for: building exhaustive digital profiles through a rigorous multi-level onboarding process. Level one captures the first name, last name, email, mobile number, password, and gender. Level two captures education, work experience, skills, training, certificates, recent salary expectations, current address, and permanent residence. Level three optionally captures social media links, personal statements, and biographies. Once authenticated, this role safely uploads resume documents for automated Artificial Intelligence parsing, manages personal privacy controls to set their profile as public or private, browses available opportunities using semantic search, receives personalized job recommendations, and saves specific job postings to customized favourite lists.

The **External Job Portal** role is granted an access level of **Restricted Application Programming Interface Integration**. Users assigned to this role must authenticate via **Strict OAuth 2.0 Cryptographic Tokens, Unique Application Programming Interface Keys, and IP Whitelisting Mechanisms**. This role is primarily responsible for: securely pushing external job postings directly into the central matching system via the RESTful Push Application Programming Interface. External portals can dynamically synchronize job status updates such as deadline extensions, description edits, or early closures using unique job identifiers assigned by the central system. This role is strictly confined to data submission and monitoring; they are provided an integration dashboard to review submission logs, track error handling responses, and monitor successful job ingestion statistics without ever gaining access to the internal job seeker database.

The **Guest User** role is granted an access level of **Unauthenticated Public Browsing**. Users assigned to this role must authenticate via **No Authentication Required**. This role is primarily responsible for: executing highly restricted basic searches utilizing broad filters such as geographic location, industry sector, and posting dates. Guest users can view public job postings, read public news articles, and browse the platform Help Center. However, to enforce data privacy and encourage platform adoption, guest users are strictly prohibited from applying for any jobs, viewing candidate profiles, or subscribing to customized alerts. The system is engineered to constantly prompt guest users to formally register and authenticate whenever they attempt to access secure platform functionalities.

# 12. State & Session Management

The management of user sessions and application states represents a paramount security vector for the Artificial Intelligence Driven Job Matching Platform. Because the system processes highly sensitive Personally Identifiable Information for vulnerable youth populations and critical corporate registration metrics, the architecture absolutely forbids implicit trust regarding active connections. The central web server dynamically tracks every single authenticated user journey, ensuring that the identity established during the initial phishing resistant login phase remains cryptographically verifiable throughout the entire session lifecycle. To completely prevent malicious actors from hijacking active sessions or exploiting dormant browser windows, the platform integrates defense mechanisms explicitly engineered to sever access the moment suspicious activities manifest.

To maintain application security and performance, user sessions are strictly managed. The system enforces a session timeout limit of **an absolute maximum of exactly fifteen minutes of idle time, combined with an explicitly configurable administrative ceiling as mandated by functional requirement FR 52**. During an active session, tokens or session data are securely stored using **highly encrypted browser cookies explicitly mandated to utilize strict HttpOnly, Secure, and SameSite Strict flags to permanently prevent unauthorized client-side script access and completely block Cross Site Request Forgery attacks**. The overall strategy for state persistence across the application is defined as: **centralized server-side session tracking managed securely by the backend architecture utilizing an encrypted in memory data store, seamlessly synchronized with continuous real time anomaly detection and strict client side cookie token validation**.

Building upon this foundation, the platform executes continuous authentication anomaly detection algorithms operating in absolute real time. These sophisticated security algorithms rigorously evaluate user behaviour, network origins, and request velocities during every single navigation event. If the anomaly detection engine identifies an unexpected geographical shift, a suspiciously rapid sequence of data extraction requests, or concurrent login attempts originating from disparate Internet Protocol addresses, the system executes an immediate Fail Closed protocol. This protocol instantly invalidates the session token, completely terminates the active connection, and permanently locks the account pending formal administrative review, guaranteeing that hijacked sessions are completely neutralized before any sensitive matching matrices or applicant resumes can be compromised.

Furthermore, the architecture completely rejects the usage of stateless client-side tokens that cannot be instantly revoked. Instead, the platform relies exclusively on cryptographically signed server-side opaque tokens. When a user requests to terminate their session, or when the fifteen minute timeout ceiling is mathematically reached, the backend infrastructure actively destroys the token record directly within the centralized encrypted memory store. This architectural decision ensures that even if an advanced persistent threat manages to perfectly clone a session cookie, the stolen token becomes completely useless the exact millisecond the backend invalidates the corresponding server-side record. By explicitly enforcing these draconian session boundaries and continuous anomaly checks, the platform guarantees that the high assurance security posture is permanently maintained from the exact moment of Multi Factor Authentication until the final logout event.

# 13. Data Flow & Boundaries

The Data Flow and Boundaries architecture establishes the absolute zero trust network perimeter and secure communication channels necessary to defend the platform against advanced persistent threats. Within this high assurance ecosystem, no single entity, internal microservice, or external Application Programming Interface request is ever implicitly trusted. To perfectly neutralize packet interception, man in the middle attacks, and unauthorized data exfiltration, the architecture explicitly dictates that all in transit communication mandates mutual TLS (mTLS) 1.3 cryptographic protocols. This rigorous requirement guarantees that both the transmitting client and the receiving server continuously and cryptographically authenticate each other during every single data exchange.

Furthermore, the strict boundary definitions ensure that the core database ledger and the isolated Artificial Intelligence execution enclaves are physically decoupled from the public internet. All external requests originating from mobile job seekers, registered corporate employers, or approved third party job portals must successfully traverse the Web Application Firewall and terminate at the secure Identity and Application Programming Interface Gateway. Once perfectly authenticated via cryptographic tokens, data is precisely routed to the appropriate internal subnet. To complete the high assurance compliance cycle, all permanent data residing within the private database enclave is explicitly secured using AES-256-GCM managed via HSM. This absolute mandate ensures that all data at rest, specifically including vulnerable Personally Identifiable

Information and spatial location coordinates, is constantly guarded by hardware level master keys that cannot be extracted by malicious software vectors.

**Data Flow Definitions:**

- When data is transmitted from the **Job Seeker or Employer Client Device** to the **Public Web Application Firewall**, the communication utilizes the **HTTPS** protocol. The expected payload type for this exchange is **JSON formatted profile submissions, resume document uploads, or job search queries**. To ensure security in transit, all data crossing this boundary is encrypted using strictly enforced **mutual TLS (mTLS) 1.3**.

- When data is transmitted from the **Secure API Gateway** to the **Isolated Artificial Intelligence Matching Enclave**, the communication utilizes the **Internal TCP/IP** protocol. The expected payload type for this exchange is **Pre sanitized text matrices requiring vector scoring and Natural Language Processing**. To ensure security in transit, all data crossing this highly sensitive internal boundary is encrypted using **mutual TLS (mTLS) 1.3**.

- When data is transmitted from the **Secure API Gateway** to the **Core System Database Management System**, the communication utilizes the **Internal TCP/IP** protocol. The expected payload type for this exchange is **Parameterized SQL statements containing relational profile data and spatial coordinates**. To ensure security in transit, all data crossing this boundary is encrypted using **mutual TLS (mTLS) 1.3**, and upon arrival, all data at rest is secured using **AES-256-GCM managed via HSM**.

- When data is transmitted from the **External Third Party Job Portals (e.g., jobs dot ps)** to the **Integration API Gateway**, the communication utilizes the **HTTPS** protocol. The expected payload type for this exchange is **Schema dot org standard job posting synchronization payloads formatted in JSON**. To ensure security in transit, all data crossing this boundary is encrypted using **mutual TLS (mTLS) 1.3** and explicitly requires OAuth 2.0 token validation.

- When data is transmitted from the **Secure API Gateway** to the **External Ministry of Labor Government Databases**, the communication utilizes the **HTTPS** protocol. The expected payload type for this exchange is **Validation requests containing company registration numbers for real time employer verification**. To ensure security in transit, all data crossing this boundary is encrypted using strictly enforced **mutual TLS (mTLS) 1.3**.

**Data Flow Diagram (DFD):**

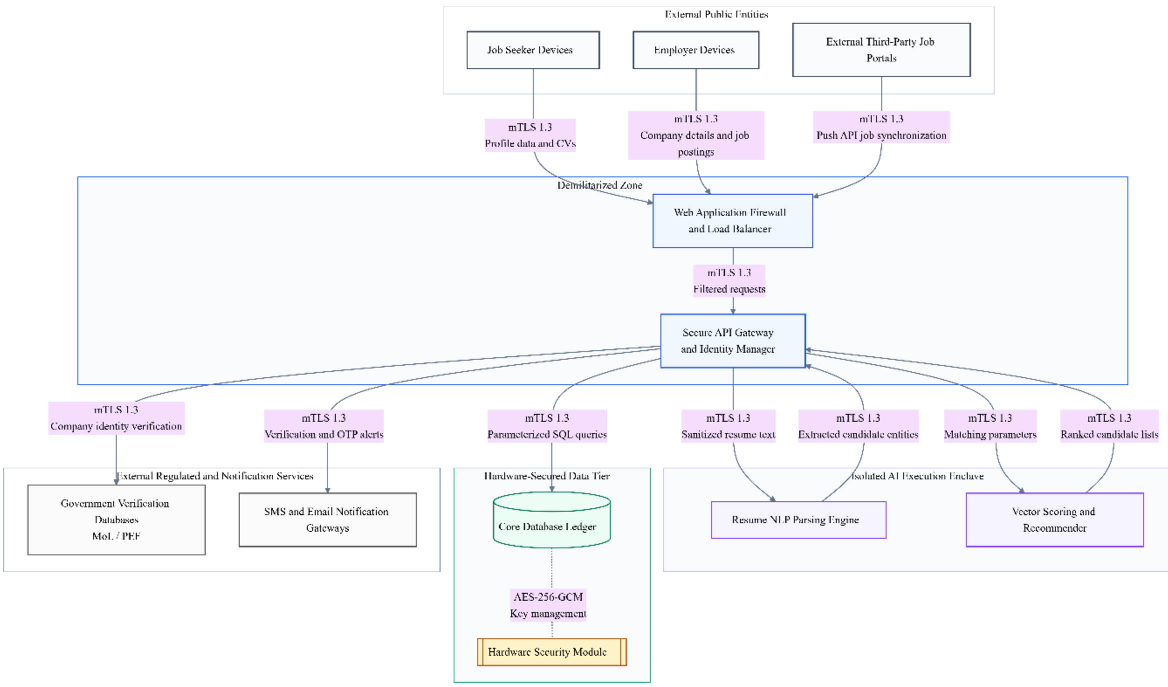


*Figure 2 AI Recruitment System Architecture and Data Exchange Diagram*

# 14. Error Handling Guidelines

The application is designed with specific error handling guidelines to prevent information leakage and ensure stability across this critical national infrastructure. Because the Artificial Intelligence Driven Job Matching Platform processes highly sensitive Palestinian youth profiles, interacts directly with Ministry of Labor databases, and calculates complex vector embeddings, any unexpected computational anomaly poses a severe security risk. Therefore, the architecture absolutely mandates that in the event of a critical failure, an integration timeout, or an ambiguous authorization check, the system defaults to a **Fail Closed** state. This explicitly guarantees that all operations are immediately halted and resources are locked down to prevent unauthorized data extraction or cascading system failures. Within production environments, stack trace visibility is explicitly set to: **Completely suppressed in all environments outside of local development**. Under absolutely no circumstances will detail system errors, database query structures, or

Artificial Intelligence algorithmic failures be exposed to end users, external job portals, or public networks.

Furthermore, the system implements standardized responses for common exceptions:

Upon encountering the **Input Validation and Malicious Payload (HTTP 400)** error, the system will execute the following handling strategy: The system will trigger an immediate Fail Closed protocol to reject the incoming data payload completely. This strategy explicitly addresses attempted prompt injections against the Artificial Intelligence matching engine and malicious script insertions. The application will log the detailed intrusion attempt securely within the administrative audit trail, alert the security operations center, and return a highly generic, sanitized validation message to the user interface without revealing the underlying parsing logic or the exact security rule triggered.

Upon encountering the **Authentication and Cryptographic Token Violation (HTTP 401 and HTTP 403)** error, the system will execute the following handling strategy: The backend security framework will instantly sever the active connection and invalidate the user session token. To neutralize brute force enumeration attacks, the system will absolutely avoid specifying whether the username, the password, or the Multi Factor Authentication component was incorrect. The system will log the anomalous access attempt, increment the automated lockout counter, and display a standardized, generic access denied message to the user while keeping the core database completely isolated.

Upon encountering the **External Integration and Government Database Timeout (HTTP 502 and HTTP 504)** error, the system will execute the following handling strategy: If an external system, such as the Ministry of Labor registration database or a third-party job portal like jobs dot ps, fails to respond within the acceptable latency threshold, the application will implement a strict circuit breaker pattern to completely prevent cascading operational failures. The platform will safely queue the pending synchronization transaction for a future automated retry, suppress all external routing logs, and present a localized notification informing the employer or job seeker that the external verification service is temporarily experiencing high latency.

Upon encountering the **Artificial Intelligence Enclave Mathematical Exception (HTTP 500)** error, the system will execute the following handling strategy: Should the Natural Language Processing components or the vector scoring algorithms experience a critical mathematical failure while processing a resume document, the isolated execution enclave will immediately and safely terminate the calculation to prevent memory corruption. All complex stack traces generated by the Artificial Intelligence framework will be entirely suppressed. The system will gracefully degrade its matchmaking functionality, notify the high-level system administrators for immediate technical investigation, and load a standard fallback interface for the job seeker.

# 15. System Telemetry & Logging

Operational excellence and security monitoring are supported through comprehensive system telemetry. All logs are aggregated using **an Enterprise Security Information and Event Management platform integrated directly with cryptographic Write Once Read Many storage arrays**. To protect sensitive information, the system employs the following PII and sensitive data redaction strategy before logs are written: **strict algorithmic scrubbing of all Personally Identifiable Information, explicitly ensuring that absolutely no sensitive youth profiles or employer national identification numbers are ever written to disk or sent to any Large Language Model node via LangChain or external Application Programming Interfaces**.

The logging strategy for the Artificial Intelligence Driven Job Matching Platform is architected to exceed standard compliance and guarantee absolute high assurance security for the Palestinian labor market. Because the platform processes extremely sensitive demographic data and executes complex vector based matching algorithms, the system telemetry must be impenetrable. To definitively prevent tampering by malicious actors or compromised internal accounts, every single log entry generated by the system is cryptographically signed the exact millisecond it is created. These signed logs are immediately routed into strictly isolated Write Once Read Many storage environments. This physical and logical hardware constraint guarantees that once an event is recorded, it mathematically cannot be altered, deleted, or manipulated by any system administrator, government official, or advanced persistent threat, ensuring an absolutely perfect forensic audit trail.

Furthermore, the telemetry architecture explicitly integrates advanced data redaction proxies at the network edge. Before any data payload is allowed to interact with the Artificial Intelligence execution enclaves, the Natural Language Processing parsers, or external monitoring dashboards, the redaction proxies actively scrub all Personally Identifiable Information. This means personal names, mobile numbers, email addresses, and exact physical locations are completely masked. This strict isolation guarantees that when user resumes are processed for skill extraction, the underlying algorithms and connected Large Language Models evaluate purely sanitized semantic text, perfectly neutralizing any risk of data leakage or privacy violations while maintaining full compliance with the General Data Protection Regulation principles.

The following core events are captured by the logging system:

- **Authentication Successes and Failures**: This encompasses: The exhaustive tracking of every single login attempt, password reset request, and Multi Factor Authentication challenge. The system securely records the originating Internet Protocol address, the physical device fingerprint, and the exact timestamp of the interaction to feed continuous anomaly detection algorithms without logging the actual password payloads.
- **Privilege Escalations and Access Control**: This encompasses: The rigorous auditing of any user or system process attempting to elevate its authorization context. This specifically includes monitoring when a standard user attempts to access the Ministry of Labor administrative dashboard, or when a verified employer attempts to bypass Attribute Based Access Control boundaries to view private candidate profiles.
- **Artificial Intelligence Pipeline Data Access**: This encompasses: The continuous monitoring of all unstructured data entering and exiting the isolated Artificial Intelligence execution enclaves. The system records the exact timestamps when resume parsing begins, the vector scoring execution times, and the mathematical confidence scores generated by the recommendation engine, ensuring total transparency of the algorithmic matchmaking process.
- **External Integration Synchronization**: This encompasses: The detailed logging of all data exchanges occurring through the Application Programming Interface gateway. This explicitly captures the synchronization payloads received from external third-party job portals like jobs dot ps, and the real time verification queries dispatched to the Palestinian Employment Fund databases, accurately recording success codes and integration timeouts.

# 16. External Dependencies & Libraries

The system relies on verified third-party components to deliver its functionality.

Because the Artificial Intelligence Driven Job Matching Platform functions as critical socioeconomic infrastructure for the Palestinian territory, the architecture absolutely refuses to implicitly trust any external software dependency. Modern web applications heavily rely on open-source frameworks, Artificial Intelligence libraries, and third-party codebases, which frequently introduce massive supply chain vulnerabilities. To completely neutralize these specific threats, the platform implements a draconian supply chain security strategy. Every single library, framework, and external package integrated into the system is subjected to continuous, automated Common Vulnerabilities and Exposures scanning during every single phase of the continuous integration and continuous deployment pipeline. If the automated scanners detect a critical vulnerability within a proposed dependency update, the deployment is instantly halted, and the security operations center is immediately alerted to initiate manual remediation protocols.

To further fortify the software supply chain and guarantee absolute transparency, the deployment architecture explicitly mandates the dynamic generation of a comprehensive Software Bill of Materials during every single application build. This continuously updated registry acts as a precise cryptographic inventory of every component, sub component, and specific library version compiled into the final production software. By maintaining this meticulous Software Bill of Materials, the Ministry of Labor and the Palestinian Employment Fund can instantly identify if the platform utilizes a newly compromised open-source library. This mechanism empowers administrators to deploy rapid security patches before malicious actors can exploit a zero-day vulnerability, ensuring the digital survivability of the platform and perfectly protecting the vulnerable youth populations relying on its matchmaking capabilities.

The application utilizes **React or Angular Enterprise Frontend Frameworks** (Version: **Latest Stable Enterprise Release**). The primary purpose of this dependency is to provide: A highly responsive, universally accessible graphical user interface that seamlessly supports right to left Arabic text rendering and strict Web Content Accessibility Guidelines compliance. The strategy for updating and maintaining this library is defined as: Automated version tracking utilizing continuous integration pipelines paired with mandatory human review of all generated Software Bill of Materials documentation prior to any production deployment.

The application utilizes **Python Artificial Intelligence and Natural Language Processing Libraries** (Version: **Latest FIPS Validated Release**). The primary purpose of this dependency is to provide: The highly complex vector-based scoring mathematics, automated resume data extraction capabilities, and semantic relationship mapping required by the core matching engine. The strategy for updating and maintaining this library is defined as: Rigorous execution within isolated testing enclaves to guarantee that version updates do not introduce algorithmic bias or degrade the mathematical precision of the job recommendations.

The application utilizes **Enterprise Relational Database Management System Engine** (Version: **Latest Long Term Support Release**). The primary purpose of this dependency is to provide: The impenetrable, highly scalable permanent storage mechanism for all encrypted user profiles, structural job postings, and relational audit logs. The strategy for updating and maintaining this library is defined as: Carefully scheduled manual maintenance windows executed strictly during non-peak operational hours, supported by complete disaster recovery snapshot generation prior to any major version upgrade.

The application utilizes **Advanced Cryptographic Key Management Libraries** (Version: **FIPS compliant and validated**). The primary purpose of this dependency is to provide: The mathematically provable encryption algorithms required to execute Advanced Encryption Standard 256-bit data at rest protection and Mutual Transport Layer Security network encryption. The strategy for updating and maintaining this library is defined as: Immediate, automated, and prioritized patching overriding all standard deployment schedules whenever critical cryptographic vulnerabilities are publicly disclosed.

# 17. Environment Configuration

Configuration is managed separately from the application code to ensure portability and security across different deployment environments. Because this platform serves as the premier digital employment infrastructure for the Palestinian workforce, environmental configurations must adhere to strict zero trust principles. The primary configuration management tool utilized is **Advanced Infrastructure as Code provisioning engines such as Terraform combined with Ansible and Kubernetes ConfigMaps**. Sensitive values, such as API keys and database credentials, are securely injected into the runtime environment using **isolated, heavily restricted Enterprise Secrets Vaults featuring automated secret rotation policies**. To guarantee absolute cryptographic security and prevent devastating data breaches, the project strictly prohibits any hardcoded credentials within the application source code or configuration files. This ensures that cryptographic keys are continuously refreshed without human intervention, rendering stolen credentials obsolete almost immediately.

Specific environment configurations are handled as follows:

For the **Production (Live National Infrastructure)** environment, variables are designated with the prefix **PROD_AIJMP_**. These configuration values are stored within: **The strictly isolated and heavily restricted Enterprise Secrets Vault located exclusively inside the private backend subnet**. This environment processes all live demographic data, sensitive youth profiles, and official employer registrations. It demands absolute compliance with the automated secret rotation mandates, ensuring that all database access credentials and external Application Programming Interface tokens are cryptographically rotated on a strict schedule without any manual administrative access.

For the **Staging and User Acceptance Testing** environment, variables are designated with the prefix **STG_AIJMP_**. These configuration values are stored within: **A segregated Staging Secrets Vault connected explicitly to sanitized and fully anonymized dataset ledgers**. This environment operates as an exact architectural clone of the production infrastructure, utilized primarily by Ministry of Labor officials and testing teams to validate new software releases. It utilizes mock endpoints for external job portals and simulated short message service gateways to guarantee that authentic notifications are never accidentally dispatched to actual citizens during testing procedures.

For the **Local Developer Sandbox** environment, variables are designated with the prefix **DEV_AIJMP_**. These configuration values are stored within: **Encrypted local configuration files that are explicitly mandated to be excluded from all centralized version control repositories**. This is the only environment where stack traces are permitted to be visible for debugging purposes. Developers utilize local host databases and simulated Artificial Intelligence containers to write and verify new code functionalities without ever touching the highly secure staging networks or viewing actual user profile metrics.

# 18. Deployment Architecture

The deployment architecture of the Artificial Intelligence Driven Job Matching Platform is meticulously engineered to provide unprecedented security, multi region high availability, and absolute protection against unauthorized external access. Because the platform operates as critical socioeconomic infrastructure for the Palestinian territory, managing highly sensitive youth demographic data and core Ministry of Labor integrations, the infrastructure absolutely mandates strict Zero Trust Network Architecture principles. The physical and logical network topology is divided into completely isolated execution tiers. The public facing Demilitarized Zone securely houses the Web Application Firewalls and enterprise load balancers to meticulously inspect all incoming Hypertext Transfer Protocol traffic. Beyond this perimeter, the application logic executes within heavily restricted application subnets.

Most critically, to definitively neutralize advanced persistent threats and prevent catastrophic data exfiltration, the architecture enforces completely isolated, air gapped private subnets for the core relational databases and the Artificial Intelligence execution enclaves. These profound data sanctuaries possess absolutely no direct internet routing capabilities. They are strictly accessible only via cryptographically authenticated mutual Transport Layer Security 1.3 connections originating exclusively from the approved internal Application Programming Interface gateways. By implementing this extreme segmentation, combined with automated containerization technologies for flawless horizontal scaling, the platform guarantees that malicious actors cannot reach the underlying vector matrices or Personally Identifiable Information even if the public facing web servers are compromised.

The system is deployed on infrastructure provided by **a Scalable High Availability Enterprise Cloud or Hybrid Infrastructure Provider**. The overall network topology and segmentation strategy is described as: **Strict Zero Trust multi-tier network segmentation establishing a**

**secure Demilitarized Zone for application routing while placing all Artificial Intelligence processing modules and central relational databases within completely isolated, air gapped private subnets**. To secure the network perimeter, the following ingress and egress controls are enforced: **Rigorous Web Application Firewall allow listing protocols that strictly reject all unauthorized incoming connections at the perimeter, combined with aggressive egress filtering that only permits the internal servers to communicate externally with explicitly approved government endpoints via mutual Transport Layer Security 1.3**.

The deployment architecture consists of the following key nodes:

- **Public Load Balancer and Web Application Firewall**: The absolute first point of contact for all incoming network traffic, responsible for terminating encrypted connections, filtering malicious payloads, mitigating distributed denial of service attacks, and intelligently routing legitimate requests to the application layer.

- **Secure API Gateway and Identity Manager**: The central authentication node operating within a restricted subnet, explicitly responsible for validating OAuth 2.0 cryptographic tokens, enforcing Role Based Access Control, and orchestrating all authorized data flows between the client interfaces and the isolated backend enclaves.

- **Artificial Intelligence Execution Enclave**: A highly secure, isolated computing cluster equipped with advanced processing units dedicated exclusively to executing the Natural Language Processing algorithms, parsing unstructured resumes, and generating high dimensional vector embeddings without exposing the models to the broader network.

- **Core Database Management System Node**: The central relational data repository locked entirely inside the air gapped private subnet, exclusively responsible for securely persisting all encrypted user profiles, job postings, and immutable audit logs.

- **External Subsystem Gateway**: The highly restricted outbound routing node utilized exclusively to securely fetch employer verification data from the Ministry of Labor and the Palestinian Employment Fund databases.

## Cloud and Network Architecture Diagram:

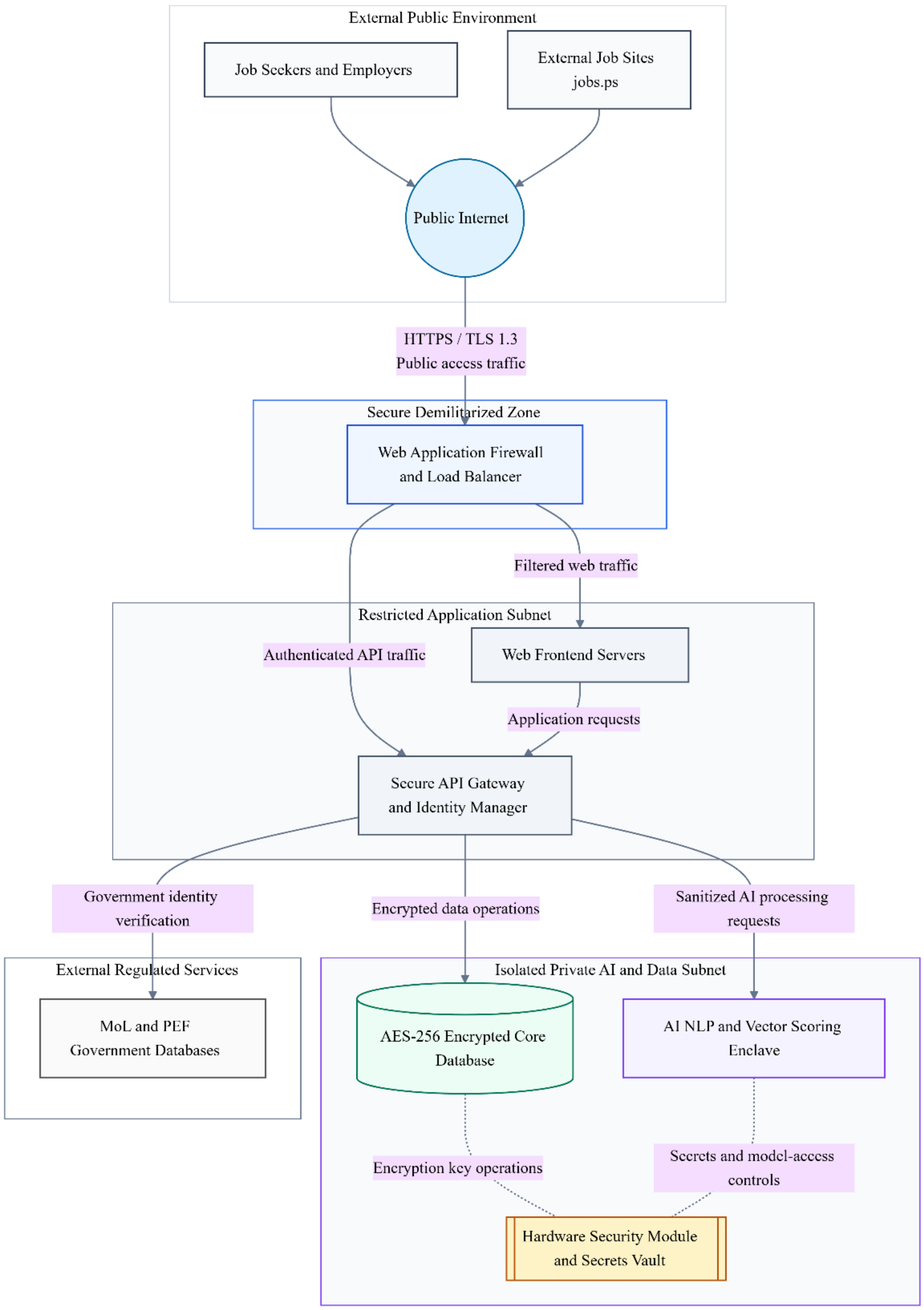


*Figure 3 Secure Network Architecture for an AI-Enabled Recruitment Platform*

# 19. Continuity & Recovery Operations

Given the paramount importance of the Artificial Intelligence Driven Job Matching Platform in facilitating daily workforce logistics and economic empowerment across the Palestinian territory, absolute data survival and rapid operational recovery are non-negotiable enterprise mandates. The platform operates as a critical socioeconomic infrastructure, meaning any prolonged outage could severely impact vulnerable job seekers and major corporate employers. To fulfill the rigorous reliability requirements established by the Ministry of Labor and the Palestinian Employment Fund, the architecture implements a highly aggressive Continuity and Recovery framework designed around multi region high availability. The platform is explicitly required to maintain a 99.5 percent availability during standard operating hours, defined precisely as 8:00 AM to 8:00 PM Palestine time from Sunday through Thursday, and a 99.0 percent availability during all nonstandard hours. To successfully achieve these stringent uptime metrics, the system implements continuous database replication to actively prevent data loss in the event of severe database failures, facility power losses, or regional network partitions.

The architecture is fundamentally engineered for extreme digital survivability and resilience. The system must automatically recover from common failure scenarios without requiring any manual administrative intervention. Furthermore, to protect the core matching engine from external bottlenecks, the platform implements sophisticated circuit breaker patterns for all external service dependencies. This explicitly prevents cascading failures if third party job portals or government verification databases experience unexpected latency or total service drops. In the event of localized internal component failures, the infrastructure guarantees that the system will continue to function with gracefully degraded performance rather than experiencing a complete systemic crash. All scheduled maintenance windows are strictly confined to periods of the absolute lowest expected system usage, and administrators must provide advance notice to all users prior to initiating these critical system updates.

To ensure business continuity in the event of a severe disruption, recovery operations are clearly defined and mathematically constrained. The target Recovery Time Objective is **strictly exactly 4 hours for all critical platform functions including user authentication, Artificial Intelligence matchmaking, and job posting, while a target of exactly 24 hours is permitted for non-critical functions such as custom report generation**. The target Recovery Point Objective is **strictly exactly 1 hour, mathematically guaranteeing that the system**

**experiences absolutely no more than one hour of potential data loss during a catastrophic disaster scenario**.

The backup strategy mandates that backups are performed at a frequency of complete relational full backups executed at least weekly, coupled with precise incremental backups executed daily. All backup data is securely stored at geographically separate locations that are completely isolated from the primary hosting infrastructure. To align with the stringent high assurance security compliance required for this critical platform, these backups utilize Write Once Read Many immutable storage arrays and are explicitly protected using Advanced Encryption Standard 256-bit cryptographic protocols to ensure the data cannot be tampered with or extracted by malicious actors.

Finally, maintaining a theoretical recovery posture is completely insufficient for critical national infrastructure. The system mandates that a highly comprehensive disaster recovery plan is permanently documented, continuously updated, and rigorously tested against emerging threat vectors. To guarantee continuous operational readiness and verify the efficacy of the automated failover mechanisms, system administrators and government officials must actively conduct and formally document exhaustive disaster recovery drills at least twice per year. If any computational errors occur during standard operations or active recovery phases, the system handles input validation gracefully, implements appropriate retry mechanisms for transient errors, and logs detailed diagnostic information while explicitly suppressing all sensitive system configurations and stack traces from the end users.

# 20. Data Lifecycle Management

The Artificial Intelligence Driven Job Matching Platform actively manages highly sensitive personal profile metrics, detailed educational histories, and proprietary corporate registration documents. Consequently, the enterprise absolutely requires rigorous Data Lifecycle Management policies to deeply protect user privacy, comply with all Palestinian data protection regulations, and adhere strictly to General Data Protection Regulation principles. The system enforces strict data minimization protocols, ensuring that demographic information and geographical coordinates are actively collected and utilized only as long as strictly necessary to fulfill the operational matching algorithms. To further empower the Palestinian workforce and guarantee digital sovereignty, the platform provides robust user mechanisms explicitly allowing job seekers and verified employers to securely view, export, and permanently delete their personal data at any time, in full accordance with modern data protection regulations.

Once a job seeker permanently deactivates their account, or once an employer successfully fills a required position and terminates the listing, the application aggressively sweeps the core database to transition this data into secured archival states or permanently destroys it. Furthermore, all system telemetry, administrative actions, and data modifications are meticulously preserved within immutable audit trails to satisfy strict compliance purposes. However, to guarantee total eradication of sensitive personal information and prevent forensic recovery by malicious state sponsored actors, the infrastructure explicitly requires stringent cryptographic erasure and digital shredding for all final data disposal events.

Standard database row deletion is mathematically insufficient for this high assurance architecture. Instead, the system mandates that the centralized Hardware Security Modules explicitly destroy the specific cryptographic keys associated with the targeted data blocks. By destroying the encryption keys rather than merely overwriting the disk sectors, the system renders the archived information permanently unreadable and absolutely unrecoverable, perfectly safeguarding the identities of vulnerable youth populations and corporate entities alike.

Data processed by the system is subject to strict lifecycle management policies to ensure compliance and optimize storage.

- Data categorized as **Job Seeker Profiles and Personally Identifiable Information** is subject to a retention period of **the exact active lifetime of the registered account, plus**

**an administrative holding period strictly defined by local Palestinian labor compliance laws**. Upon reaching the end of this period, the archival strategy employed is **immediate transfer to a heavily encrypted cold storage ledger strictly for legal audit resolution and regulatory compliance**. Once the data is no longer required for archival purposes, it is securely destroyed using the following disposal method: **Rigorous cryptographic erasure and permanent digital shredding of the storage media blocks to render all personal biographies, email addresses, and mobile numbers mathematically unrecoverable**.

- Data categorized as **Employer Registration Records and Government Verification Metrics** is subject to a retention period of **the entire duration of the active corporate account engagement on the platform**. Upon reaching the end of this period, the archival strategy employed is **complete anonymization of hiring trends and secure vault storage of official Ministry of Labor verification badges for historical auditing purposes**. Once the data is no longer required for archival purposes, it is securely destroyed using the following disposal method: **Hardware level cryptographic shredding and secure destruction of all associated verification certificates and uploaded supplementary company documents**.

- Data categorized as **Artificial Intelligence Vector Embeddings and Parsed Resume Matrices** is subject to a retention period of **exactly the duration the associated resume or job posting remains active and visible within the central matching engine**. Upon reaching the end of this period, the archival strategy employed is **complete mathematical anonymization to extract purely statistical labour market trends while irreversibly decoupling the semantic data from the originating user identity**. Once the data is no longer required for archival purposes, it is securely destroyed using the following disposal method: **Immediate cryptographic erasure directly from the isolated Artificial Intelligence execution enclave memory and backend vector database**.

- Data categorized as **System Telemetry and Immutable Security Audit Logs** is subject to a retention period of **a minimum of exactly five years to satisfy stringent national security, legal compliance, and accountability mandates**. Upon reaching the end of this period, the archival strategy employed is **secure preservation within geographically isolated Write Once Read Many storage arrays**. Once the data is no longer required for archival purposes, it is securely destroyed using the following disposal method: **Mandatory cryptographic shredding coordinated with authorized physical hardware destruction protocols officially witnessed by Ministry of Labor compliance officials**.

# 21. Assumptions and Constraints

**Assumptions:**

- **Geographic Network Resilience and Telecommunication Stability:** Given the unique geopolitical realities and physical infrastructure challenges present within the Palestinian territory, the system architecture inherently assumes that core telecommunication networks connecting the West Bank, Gaza, and East Jerusalem might experience transient instability or sudden bandwidth degradation. However, it is assumed that a minimum baseline of connectivity will remain active to support the rigorous mutual Transport Layer Security 1.3 cryptographic handshakes required for the secure authentication of job seekers and employers.

- **Government Database Interoperability and Continuous Availability:** The architectural design strictly assumes that the Palestinian Ministry of Labor and the Palestinian Employment Fund will consistently maintain the uptime and secure accessibility of their internal verification databases. The platform relies on the assumption that these external government entities will provide timely, uninterrupted Application Programming Interface access to facilitate the real time verification of corporate registration credentials and youth educational certificates without unexpected schema deprecations.

- **Phased Implementation and Scalability Trajectory:** It is assumed that the software development lifecycle will follow a strictly phased implementation approach, prioritizing the deployment of the core matchmaking engine and secure user onboarding functionalities prior to expanding the system into a comprehensive, overarching Labor Market Information System. The architecture assumes a baseline capacity scaling capable of supporting a one hundred percent annual growth in the user base for at least the first three years of active national operation.

- **External Integration Cooperation:** The project explicitly assumes that highly utilized third party regional job portals, specifically platforms such as jobs dot ps, will actively cooperate with the integration mandates. It is assumed these external entities will adopt the provided secure RESTful Push Application Programming Interface specifications and align their data structures with the required Schema dot org JobPosting standards to guarantee flawless semantic synchronization.

- **End User Technological Literacy and Device Availability:** The deployment strategy assumes that the targeted demographic of Palestinian youth, specifically those aged fifteen to twenty-nine, possess adequate digital literacy and access to standard client devices such as smartphones or laptops equipped with modern, updated web browsers capable of executing complex responsive graphical user interfaces and secure hardware token authentication.

**Constraints:**

- **Absolute High Assurance Security and Digital Survivability:** Because this platform serves as the premier socioeconomic infrastructure for vulnerable youth populations, it is explicitly constrained by stringent Open Worldwide Application Security Project compliance mandates. The system is fundamentally restricted from implicitly trusting any internal or external network request. It must operate within a strict Zero Trust Network Architecture, utilizing completely isolated, air gapped private subnets for all core database ledgers and Artificial Intelligence execution enclaves to ensure absolute protection against advanced persistent threats and malicious state sponsored data exfiltration vectors.

- **Artificial Intelligence Algorithmic Boundaries and Decision Support Limitation:** To guarantee algorithmic fairness and prevent automated employment bias, the Artificial Intelligence engine is strictly constrained to operating exclusively as a decision support and recommendation mechanism. The Natural Language Processing parsers and vector scoring models are explicitly prohibited from possessing autonomous hiring authority. The system mathematically cannot execute automatic candidate rejections on behalf of an employer and must continuously restrict its output to generating ranked shortlists and extracted skill matrices for manual human review.

- **Rigorous Data Privacy and Regulatory Compliance:** The platform is severely constrained by strict adherence to General Data Protection Regulation principles and all applicable Palestinian national data privacy laws. The architecture is legally required to implement absolute data minimization, ensuring Personally Identifiable Information is never exposed to unauthorized entities. Furthermore, the system is constrained by the absolute necessity to provide job seekers with fully automated mechanisms to view, export, and permanently destroy their personal data through rigorous cryptographic shredding methodologies.

- **Disaster Recovery and Operational Continuity Limits:** The continuous availability of the matchmaking platform is constrained by aggressive digital survivability metrics. The infrastructure must definitively guarantee a Recovery Time Objective of exactly four hours for all critical authentication and matchmaking functions, coupled with a Recovery Point Objective of exactly one hour. These constraints mandate continuous, highly encrypted database replication to geographically separate Write Once Read Many storage arrays, permanently preventing any localized hardware failure from disrupting the national labor market.
- **Multilingual Semantic Parity and Accessibility Mandates:** The visual presentation layer and the underlying Natural Language Processing algorithms are constrained by the absolute requirement to flawlessly support both the Arabic and English languages with perfect semantic parity. The system must seamlessly execute right to left text rendering without architectural latency and must strictly comply with Web Content Accessibility Guidelines version 2.1 Level AA standards to guarantee equitable access for all Palestinian users, including individuals relying entirely on specialized assistive technologies

## 22. Glossary

| Term | Definition |
|---|---|
| Advanced Encryption Standard 256 | A mathematically provable, government grade cryptographic algorithm utilizing a two hundred- and fifty-six-bit key length, strictly mandated across this architecture to completely secure all permanent data at rest, including user profiles and vector matrices, against unauthorized physical or digital extraction. |
| Application Programming Interface | A meticulously defined set of secure communication protocols and routing endpoints utilized by the central web server to seamlessly interface with external subsystems, explicitly requiring OAuth 2.0 cryptographic tokens and mutual Transport Layer Security 1.3 tunneling for all data exchanges. |
| Artificial Intelligence | Advanced computational frameworks and execution enclaves utilized within the platform to perform highly complex automated resume parsing, semantic intent understanding, and personalized job recommendations without requiring manual human keyword analysis. |
| Attribute Based Access Control | An advanced enterprise authorization paradigm that dynamically evaluates specific granular user attributes, geographic locations, and government verification statuses to grant or permanently deny access to secure data modification endpoints. |
| Enabel | The Belgian agency for international cooperation, acting as a primary stakeholder, donor, and overseer of the Youth Economic Empowerment in Palestine project, responsible for approving the foundational system requirements alongside local government bodies. |
| Fail Closed | A mandatory high assurance enterprise security principle ensuring that whenever an authorization check fails, an anomaly is detected, or a critical mathematical error occurs, the application absolutely defaults to denying access and instantly locking down all secure resources. |
| General Data Protection Regulation | A comprehensive international data privacy framework explicitly adopted as the foundational best practice standard for this platform, mandating strict user consent protocols, data minimization, and the absolute right to be forgotten via cryptographic erasure. |
| Hardware Security Module | A dedicated, physically isolated cryptographic hardware appliance strictly utilized to securely generate, heavily protect, and automatically rotate the master encryption keys required for database protection, |

| | |
|---|---|
| | permanently preventing administrative credential theft. |
| Labor Market Information System | The comprehensive, overarching national digital infrastructure planned for the Palestinian territory, designed to integrate multiple job sources, government databases, and economic analytics, with this job matching platform serving as its central operational nucleus. |
| Ministry of Labor | The primary Palestinian governmental entity serving as the core solution owner, responsible for reviewing requirements, providing official administrative system management, and authorizing the verification badges for private sector corporate employers. |
| Multi Factor Authentication | A rigorous identity verification protocol requiring all users to provide a secondary proof of physical identity, explicitly mandating phishing resistant hardware keys or biometric authenticators for administrative roles to prevent unauthorized account breaches. |
| Mutual Transport Layer Security 1.3 | The mandated cryptographic communication protocol strictly utilized to perfectly encrypt all data payloads traversing the network, requiring both the transmitting client and the receiving server to continuously authenticate each other during every single data exchange. |
| Natural Language Processing | A specialized branch of Artificial Intelligence explicitly deployed within the isolated execution enclaves to accurately comprehend the deep semantic meaning, contextual relationships, and unstructured skills embedded within uploaded resumes and published job descriptions. |
| Palestinian Employment Fund | The national organization acting in partnership with the Ministry of Labor, operating as a primary stakeholder and reviewer of the platform requirements, directly involved in tracking employment statistics and empowering the regional workforce. |
| Prompt Injection | A highly sophisticated Artificial Intelligence specific cyberattack where malicious actors embed hidden instructions within unstructured text, strictly neutralized within this architecture by aggressive data sanitization agents located prior to the natural language processing enclaves. |
| Recovery Point Objective | The strictly defined maximum acceptable amount of data loss measured chronologically, explicitly mandated as exactly one hour to guarantee that the system perfectly recovers from catastrophic disasters without significant national transaction loss. |
| Recovery Time Objective | The strictly targeted duration of time within which the core authentication and matchmaking business processes must be |

| | |
|---|---|
| | completely restored following a severe disruption, mathematically mandated as exactly four hours for this critical infrastructure. |
| Retrieval Augmented Generation Context Poisoning | An advanced vector attack vector where malicious adversarial inputs are intentionally uploaded to corrupt the semantic matching models, explicitly prevented by strict physical and logical quarantine of the Artificial Intelligence enclave. |
| Role Based Access Control | A rigorous security framework that heavily restricts application access and graphical interface visibility based on the specific operational roles assigned to users, explicitly preventing unauthorized privilege escalation between job seekers and employers. |
| Schema dot org JobPosting | An internationally recognized, mathematically consistent data structuring standard strictly utilized by the platform to guarantee absolute semantic compatibility across all internal job listings and synchronized external third-party integration portals. |
| Software Requirements Specification | The comprehensive foundational document detailing every functional capability, operational boundary, and technical constraint governing the entire software project lifecycle, utilized as the blueprint for the digital infrastructure. |
| Terms of Reference | The official procurement and project definition document outlining the scope, objectives, and deliverable expectations for the Phase Two system implementation process, directly supported by the requirements captured within this technical specification. |
| Vector Embedding | A highly complex, high dimensional mathematical representation of user skills and job descriptions generated by the Artificial Intelligence engine to calculate precise match percentages and power the personalized recommendation algorithms. |
| Web Content Accessibility Guidelines | An internationally recognized set of digital inclusivity standards, specifically mandated at version 2.1 Level AA, explicitly guaranteeing that individuals with disabilities can safely and effectively navigate the entire matchmaking platform. |
| Write Once Read Many | A strict physical and logical hardware storage constraint utilized for the permanent preservation of system telemetry and security audit logs, absolutely guaranteeing that recorded events mathematically cannot be altered, deleted, or manipulated by any entity. |
| Youth Economic Empowerment in | The overarching national socio-economic program aiming to drastically enhance the employability, digital inclusion, and economic |

| | |
|---|---|
| Palestine | independence of Palestinian youth, functioning as the primary funding and operational catalyst for this software project. |
| Zero Trust Architecture | An extreme enterprise security paradigm explicitly assuming the operational network is constantly hostile, dictating that absolutely no implicit trust is granted to any user or microservice, requiring continuous cryptographic validation for every internal process. |

# 23. Approval

**Approved By:** Ministry of Labor (MoL), Palestinian Employment Fund (PEF), and Enabel
**Approval Date:** May 19th, 2025